UNDERGRADUATE PROGRAMME UNDER NEP 2020
B.A. GEOGRAPHY/B.A.GEOGRAPHY (HONOURS)/B.A.GEOGRAPHY (HONOURS WITH RESEARCH), SEMESTER I & II
PANJAB UNIVERSITY, CHANDIGARH

YEAR 1 (UG CERTIFICATE)				
CourseType	**CourseCode**	**CourseTitle**	**Marks**	**Credit**
SEMESTER-I				
DisciplineSpecificCore Course (DSC 1)	GEO-DSC	Introduction to Geography	100	4
MinorDisciplinary Course (MDC 1)		To be chosen from the given basket of subjects offered	100	4
Minor Disciplinary Course (MDC 2)		To be chosen from the given basket of subjects offered	100	4
Multi-disciplinary Course(MDSC 1)	GEO-MDSC	Geography and Environment	75	3
AbilityEnhancement Course (AEC 1)		English/Modern Indian Language (MIL)	50	2
SkillEnhancement Course/Internship/ Dissertation (SEC 1)			75	3
Value Added Course (VAC 1)			50	2
Compulsory Course (Paper 1)		Punjabi/History & Culture of Punjab	50	2
		Total:	***600***	***24***
SEMESTER-II				
DisciplineSpecificCore Course (DSC 2)	GEO-DSC	Geomorphology	100	4
Minor Disciplinary Course (MDC 1)		To be chosen from the given basket of subjects offered	100	4
Minor Disciplinary Course (MDC 2)		To be chosen from the given basket of subjects offered	100	4
Multi-disciplinary Course (MDSC 2)	GEO-MDSC	Fundamentals of Disaster Management	75	3
AbilityEnhancement Course (AEC 2)		English/Modern Indian Language (MIL)	50	2
SkillEnhancement Course/Internship/ Dissertation (SEC 2)			75	3
Value Added Course (VAC 2)			50	2
Compulsory Course (Paper 2)		Punjabi/History & Culture of Punjab	50	2
		Total:	***600***	***24***
Under Graduate Certificate Awarded with Cumulative Credits				***48***

Student exiting the programme after securing 48 credits will be awarded ug certificate in the relevant discipline/subject provided they secure 4 credits in work based vocational courses offered during summer term or internship/apprenticeship in addition to 6 credits from skill-based courses earned during first and second semester.

Note:

- Students will be studying three different subjects (4 credits each). The students will finalise at the end of 2nd semester one major subject and one minor subject, to complete his/her degree in 3rd year/4th year.
- However, a student can also choose even 2 subjects in the ratio of 2:1. For example, a candidate can choose 2 papers of Geography and 1 Paper of Sociology resulting into 3 papers out of 2 subjects.
- If student has opted for a 2-credit course of a particular language, he/she can opt for another course of 3 or 4 credits in the same language. However, another 2-credit course of same language in one semester is not permitted.

SEMESTER I

COURSE TITLE:GEOGRAPHY AND ENVIRONMENT

Course Code: **GEO-MDSC**
Typology of Course: Multi-disciplinary Course (MDSC-1)
Level of Course: **100-199**
Total Credits: **3(Theory 2+ Tutorial 1)**
Maximum Marks: **75 (68 Theory + 7 Internal Assessment)**
Course Duration: **Lectures- 30 Hours; Tutorial- 15 Hours**

Evaluation: Maximum Marks: **75 (68 Theory + 7 Internal Assessment)**

- Terminal Examination: 68 Marks
- Internal Assessment: 7 Marks
- Examination Time: 3 Hours

COURSE OBJECTIVES: The course aims at developing environmental sensitivity among students to inculcate a strong sense of environmental conservation and management. The course is designed to and generate awareness about environmental challenges humanity is facing and place these issues within the frame sustainable development.

COURSE OUTCOMES: The course focuses on following outcomes:

- Comprehend key environmental processes, interactions and linkages
- Evaluate human-environment associations and issues of ecological stability
- Measure human induced changes of the environment and resultant challenges
- Critical appraisal of global sustainability initiatives

PEDAGOGY: There must be active interaction between teacher and students on different conceptual and methodological aspects of ecology and environment. The emphasis should be given to group reading, participatory learning, analysis of documentaries and discussions on complexity of key concepts and ideas of ecosystem approach to solve environmental problems with specific focus on India and local areas.

INSTRUCTIONS FOR PAPER SETTER AND CANDIDATES:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (Total 7x2= 14 marks).
2. A total of six questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *three*questions selecting one from each unit and each question will carry 18 marks (Total 3x18= 54 marks). These will be in addition to the compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (4) in the question paper.

COURSE CONTENTS

Unit I: Introduction to Environment

1. Key Concepts: Environment,Ecology, and Spheres of Earth's Environment
2. Nature and Scope of Environmental Geography
3. Ecosystem Approach: Components and types

Unit II: Environmental Pollution and Concerns

4. Environmental Pollution: Air, Water and Soil
5. Environment and energy Resources: Non-renewable and Renewable
6. Population and Environment Interface; Climate Change Issues

Unit III: Environmental Sustainability and Issues

7. Sustainable Developmentapproach; Ecological and Carbon footprints
8. Biodiversity Loss and extinction; Conservation of biodiversity
9. Environmental conservation and management: Approaches and strategies

SUGGESTED READINGS

1. Allison, S.K. and Stephen D. Murphy. (2017).*Routledge Handbook of Ecological and Environmental Restoration.* Routledge.
2. Brar, K. K. (1999).*Green Revolution: Ecological Implications*. Dominant Pub., Delhi.
3. Castree, N. and Demeritt, D. Liverman, D. and Rhoads, B. (2016). *A Companion to Environmental Geography.* Wiley, Germany.
4. Duram, L. A. (2018). *Environmental Geography: People and the Environment*. United States: ABC-CLIO.
5. Ellis, Erle C., Kees Klein Goldewijk, Stefan Siebert, Deborah Lightman and Navin Ramankutty. (2010).Anthropogenic transformation of the biomes, 1700 to 2000. *Global Ecology and Biogeography.*19: 589–606.
6. MacDonald, G. (2003).*Biogeography: Introduction to Space, Time, and Life.* John Wiley, New York.
7. Mathur, H.S. (1988).*Essentials of Biogeography.* Pointer Publishers, Jaipur.
8. Moseley, W. G., Hapke, H. M., Perramond, E., and Laris, P. (2013). *An Introduction to Human-Environment Geography: Local Dynamics and Global Processes*. Germany: Wiley.
9. Odum, E.P. and Barrett, G.W. (2009).*Fundamentals of Ecology.* 5th edition, Cengage Learning India Private Limited, New Delhi.
10. Robinson, H. (1982).*Biogeography.* The English Language Book Society and Macdonald and Evans, London.
11. Singh, Savindra (2018).*Environmental Geography.* Pravalika Publications, Allahabad.
12. Steffen, W., Jacques Grinevals, Paul Crutzen and J. McNeil (2011).The Anthropocene: Conceptual and Historical Perspectives. *Philosophical Transactions of the Royal Society. A*. 369: 842-867.
13. Tivy, Joy. (1993). *Biogeography:A Study of Plants in the Ecosphere.* Longman Scientific & Technical, U.K.

SEMESTER II

COURSE TITLE: FUNDAMENTALS OF DISASTER MANAGEMENT

Course Code: **GEO-MDSC**
Typology of Course: Multi-disciplinary Course (MDSC-2)
Level of Course: **100-199**
Total Credits: **3 (Theory 2+ Tutorial 1)**
Maximum Marks: **75 (68 Theory + 7 Internal Assessment)**
Course Duration: **Lectures- 30 Hours; Tutorial- 15 Hours**

Evaluation: Maximum Marks: **75 (68 Theory + 7 Internal Assessment)**

- Terminal Examination: 68 Marks
- Internal Assessment: 7 Marks
- Examination Time: 3 Hours

COURSE OBJECTIVES: This course aims at introducing the students with the idea of natural hazards and disaster management. It focuses on developing awareness about forms, mechanisms and processes of hazards and developing skills to analyse vulnerability, risk reduction and resilience.

COURSE OUTCOMES: At the end of this course, the students will be able to:

- Evaluate hazard and disaster mechanisms
- Determine disaster resilience and design mechanisms for safer society
- Critical evaluation of disaster management plans and policies

PEDAGOGY: Teaching and learning must be based on interaction between teacher and students on different aspects of natural hazards and disasters with the help of movies/documentaries, pictures and case studies. The emphasis will be given on group discussion and activities to promote learning on different facets of disaster risk, resilience and management.

INSTRUCTIONS FOR PAPER SETTER AND CANDIDATES:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (Total 7x2= 14 marks).
2. A total of six questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *three* questions selecting one from each unit and each question will carry 18 marks (Total 3x18= 54 marks). These will be in addition to the compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (4) in the question paper.

COURSE CONTENTS

Unit I: Basic Concepts

1. Basic Concepts: Hazard, Disaster
2. Key Concepts: Vulnerability, Risk
3. Classification/Types of Hazards and Disasters

Unit II: Disaster Management Initiatives

4. Disaster Management: Pre-disaster; Emergency and Post-disaster phase
5. Hyogo Framework (HFA) for Action
6. Sendai Framework for Disaster Management

Unit-III: Disasters and Disaster Management in India

7. Geography of Hazards/Disasters in India.
8. Disaster Management Policy of India
9. CBDRM; Gender Sensitivity in Disaster Management

SUGGESTED READINGS

1. Alexander, D. E. (2000). *Confronting Catastrophe: New Perspectives on Natural Disasters*. Harpenden, U.K: Terra Publishing.
2. Birkmann, J. (2006). *Measuring Vulnerability to Natural Hazards: Towards Disaster Resilient Societies*. US: United Nations University Press.
3. Blaikie, P. and Others (1994). *At Risk: Natural Hazards, People's Vulnerability, and Disasters,* Routledge, London.
4. Burton, I., Kates, R.W. and White, G.F. (1993). *Environment as Hazard*, 2nd edition, Guilford Press, New York.
5. Coppola, Damon P. (2015). *Introduction to International Disaster Management*. Third Edition, Elsevier, London.
6. Gupta, M.C.; L.C. Gupta, B.K. Tamini, Vinod K. Sharma (2000). *Manual on Natural Disaster Management in India,* National *Disaster Management Centre, New Delhi*
7. Hewitt, K. (1997). *Regions of Risk: A Geographical Introduction to Disasters*, Longman, London.
8. Hyndman, D. and D. Hyndman. (2009). *Natural Hazards and Disasters*. 2nd edition. USA, Belmont: Brooks/Cole.
9. Kasperson, J.X., Kasperson, R.E. and Turner, B. L. (1995). *Regions at Risk: Comparisons of Threatened Environments, United Nation* University Press, Tokyo.
10. Schneider, S.K. (1995). *Flirting with Disaster: Public Management in Crisis Situations*, M.E. Sharpe, New York